Modules Information	
Module	**HCI 3202 HUMAN COMPUTER INTERFACE**
Session	JUN 2026
Programme	BITI : Bachelor of Information Technology
Lecturer	**KALA DEVI MANAGURAN** kaladevi.managuran@newinti.edu.my Phone: 06-7982 000 Room: 3A3-11

Summary of Coursework Breakdown

(as stated in course structure)

No	Description of coursework	Learning Outcomes covered	Marks allocated
1	Online Assessment	CLO2	20%
2	Group Assignment	CLO3	40%
		Contribution of the coursework to the course	**60%**
		Final Assessment	**40%**
		Total	100%

Penalty for late submission

1 day – minus 20% of total mark awarded.
2 days – minus 50% of total mark awarded.
3 days – 0 mark for this piece of coursework.

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GROUP ASSIGNMENT			
Module	**HCI 3202 HUMAN COMPUTER INTERFACE**		
Session	**JUN2026**		
Programme	**BITI: BACHELOR OF INFORMATION TECHNOLOGY**		
Section	□ **6H1 / 6MB1**		
Lecturer/Examiner	**Kala Devi Managuran**		
Coursework Type	**Group Assignment**		
Percentage	**40% out of 100%**		
Hand-out Date	**Week 5**	Due Date	**Week 7**

Student's Declaration

We declare that:

- We understand what is meant by plagiarism, self-plagiarized and academic dishonesty.
- This project is all our own work and we have acknowledged any use of the published or unpublished works of other people.
- We hold a copy of this assignment which we can produce if the original is lost or damaged.

No.	Name	ID	Signature	Date
1	Shahadat Hossain	I26041279	Shahadat Hossain	17/7/2026
2	Rowan Cesar Macuacua	I25037217	Rowan Cesar	17/7/2026
3	Raduan Mia	I25035605	Raduan Mia	17/7/2026
4	Morris Kinyua	I24028729	Morris Kinyua	17/7/2026

Learning Outcomes Accessed

CLO3	Integrate ethics and professionalism while conducting data collection (A4, PLO11).

Student name (no.)	#1	#2	#3	#4
Report (80)				
Individual Presentation (20)				
Total (100)				
Penalty				
Final Marks (40%)				

For Lecturer:

Received by: _____ Date: _____

( )

Lecturer's Comments
Penalty for late submission:
1 day – Minus 20% of total marks awarded. 2 days – Minus 50% of total marks awarded. 3 days – 0 mark for the piece of coursework.

Note: Plagiarism & Copying

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ASSESSMENT CRITERIA AND GRADING SHEET

Task No.	*Breakdown of assessment criteria*	*Weight (%)*	*Range of achievement (✓)*					*Marks Awarded (%)*
			Excellent	*Above Average*	*Satisfactory*	*Below Average*	*Poor*	
1)	**Section A: Report**	**80%**						
	Introduction	10	10	8	6	4	2	
	Justification of Current Interface	10	10	8	6	4	2	
	Data Gathering Techniques	10	10	8	6	4	2	
	Target Users, Tasks and Sub-tasks	10	10	8	6	4	2	
	Evaluation	10	10	8	6	4	2	
	Interaction Design - Low Fidelity prototype - Justification on prototype design	20	20	16	12	8	4	
	Conclusion and References	10	10	8	6	4	2	
	TOTAL	**80%**						

2)	Section B: Individual Presentation	20%	

Student #1 Name:	Excellent	Above Average	Satisfactory	Below Average	Poor	Your Score
Content	10 ☐ Easy to follow and is complete. ☐ Presented extremely well. ☐User Research and Design choices clearly explained.	8 ☐ Structured neatly and well. ☐Easy to follow. ☐Clearly explains User Research that was conducted.	6 ☐Presentation flows in somewhat logical manner from beginning to end.	4 ☐Presentation is hard to follow.	2 ☐Unable to presentation.	
Presentation skills	10 ☐Maintains good eye contact with the audience and is appropriately animated (e.g., gestures, expression, etc.). ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice ☐Good language skills and pronunciation.	8 ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	6 ☐ Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	4 ☐ Poor language skills and poor pronunciation – cannot be understood.	2 ☐Failed To understand and present.	
Subtotal						

Student #2 Name:	Excellent	Above Average	Satisfactory	Below Average	Poor	Your Score
Content	10 ☐ Easy to follow and is complete. ☐ Presented extremely well. ☐User Research and Design choices clearly explained.	8 ☐ Structured neatly and well. ☐Easy to follow. ☐Clearly explains User Research that was conducted.	6 ☐Presentation flows in somewhat logical manner from beginning to end.	4 ☐Presentation is hard to follow.	2 ☐Unable to presentation.	
Presentation skills	10 ☐Maintains good eye contact with the audience and is appropriately animated (e.g., gestures, expression, etc.). ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice ☐Good language skills and pronunciation.	8 ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	6 ☐ Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	4 ☐ Poor language skills and poor pronunciation – cannot be understood.	2 ☐Failed To understand and present.	
Subtotal						

Student #3 Name:	**Excellent**	**Above Average**	**Satisfactory**	**Below Average**	**Poor**	**Your Score**
Content	10 ☐ Easy to follow and is complete. ☐ Presented extremely well. ☐User Research and Design choices clearly explained.	8 ☐ Structured neatly and well. ☐Easy to follow. ☐Clearly explains User Research that was conducted.	6 ☐Presentation flows in somewhat logical manner from beginning to end.	4 ☐Presentation is hard to follow.	2 ☐Unable to presentation.	
Presentation skills	10 ☐Maintains good eye contact with the audience and is appropriately animated (e.g., gestures, expression, etc.). ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice ☐Good language skills and pronunciation.	8 ☐Delivery is poised, controlled and smooth. ☐Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	6 ☐ Clear, audible voice. ☐Language skills and pronunciation can be followed and understood.	4 ☐ Poor language skills and poor pronunciation – cannot be understood.	2 ☐Failed To understand and present.	
Subtotal						

Table of Contents

1.0 Introduction 08
2.0 Justification of Current Interface 08
3.0 Data Gathering Techniques 09
4.0 Target Users, Tasks and Sub-tasks 10
5.0 Evaluation 11
6.0 Interaction Design 12
6.1 Prototypes 13
6.2 Justification on Prototype Design 17
7.0 Conclusion and References 17
Appendices 19

1.0 Introduction

The current system of using a vending machine at INTI International University, Nilai is still manual (press a button, insert a coin) where there is no guidance flow, no feedback or intelligent machine. It results in confusion regarding pricing, payment and product availability, particularly with the increased adoption of cashless payments with regard to the payment system landscape in Malaysia (Aljohi et al., 2023). When the SSTs are poorly designed, they lead to higher user errors and a lower level of satisfaction for the user, while high levels of satisfaction and low error rates are achieved when the SSTs are better designed as they are used to provide a high level of independence for users and efficiency (Ayob and Baba, 2025; Nielsen, 2020).

In this report, these usability concerns are explored and a Digital Touchscreen UI is proposed that consists of eight screens: Main Dashboard, Product Select/Detail, Add to Cart Confirmation, Cart Page, Checkout Page, Payment Method Selection, Payment Declined/Failed, and Order Confirmed. The redesign was informed by a Google Forms survey (n = 43 INTI students) which found that they were in need of the redesign and analysed in **Sections 3.0** and **5.0**.

The rest of the report is organized as follows: In **Section 2.0** the motivations for the deficiencies of the current interface are explained, and in **Section 3.0**, techniques for collecting and utilizing the data are described. Target Users and Tasks are described in **Section 4.0**, the survey findings are evaluated in **Section 5.0** and prototypes and design justifications presented in **Section 6.0**, with references presented in **Section 7.0.**

2.0 Justification of Current Interface

The present-day vending machine is completely dependent on the physical pressing of its buttons and on the manual inserting of coins, and does not have an LCD screen to check stock, prices, and payment. This design poses several issues with INTI students.

The survey results indicate that this is correct -- **60.5%** said that they had a problem with using the machine, with the most frequent response being that it is hard to understand how to pay in cash (**53.5%**) followed by not dispensing products if they paid for them (**14%**) and unclear instructions (**14%**). On the other hand, only **39.5%**

agreed that the current instructions were clear and **58.1%** were neutral, suggesting there is possibly a lack of effective communication on the process in the current interface design.

These findings support literature in HCI which states that one of the fundamentals of usability is that self-service machines do not meet a key rule of the "heurger") and users cannot be sure whether what they have performed (e.g., payment, selection) has been registered (Nielsen, 2020). Likewise, kiosk research on interaction demonstrates that ambiguous feedback and weilpomtach buttons have a substantial impact on user confidence and a higher tendency to abandon transactions (Kang et al., 2023). The absence of a digital interface also means that users with accessibility needs are not included: visual and step-based instructions have the highest impact for such users (Kang et al., 2023).

This makes it imperative to redesign this manual, touchscreen technology tool that would give immediate feedback, clear instructions, and easy payment processing.

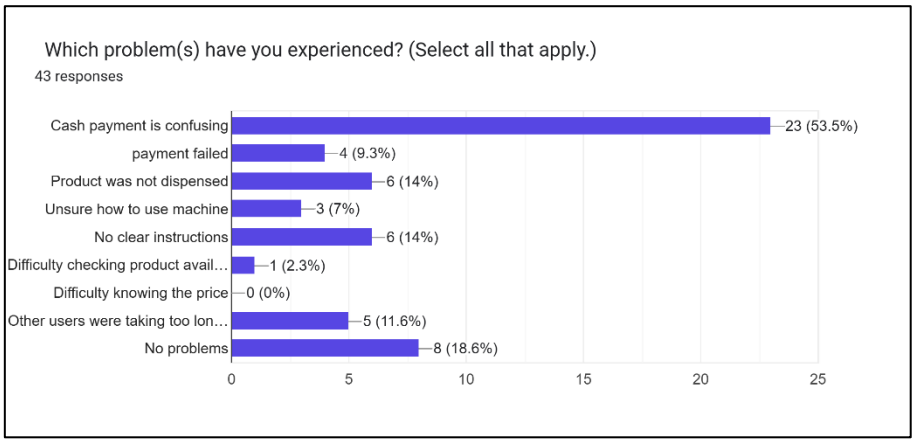


Figure 1: Problems experienced by students while using the current vending machine (n = 43)

3.0 Data Gathering Techniques

Structured online survey was identified as the main method of data collection and Google Forms was used to gather information from the students of **INTI International University**. The use of this method was because it can get a large sample within a short time, the process of answering the questionnaire can be discreet, and it can produce data that can be quantified and used for analysis of usability.

A **43 valid** response was obtained from the survey, and it was intended to be answering closed-ended questions (multiple-choice, checkbox, 5-point Likert scale)

about usage frequency, encountered problems, clarity of instructions, payment misunderstanding, satisfaction level and preferences for a proposed digital UI. Subjective ease-of-use and satisfaction were measured with Likert-scale questions on a standardized scale, as is common practice in **HCI** (Nielsen, 2020).

To help support the quantitative approach this was complimented by an open-ended "suggestions" question, which enabled the students to describe specific pain points or features, that further helped support the findings from the closed questions.

4.0 Target Users, Tasks and Sub-tasks

Target Users: Undergraduate and exchange students of the primal users of the vending machine (VM), mainly from INTI International University; and secondly, lecturers and staffs who uses the VM after lessons, mainly from INTI International University. Survey data indicates **55.8%** use the machine daily, evidencing the high frequency/time sensitive user base.

Primary Task: Use the vending machine confidently to buy a product from it.

Sub-tasks:

1. Browse available products (Main Dashboard)
2. Select a product and view details/price (Product Select/Detail)
3. Confirm the item before purchase (Add to Cart Confirmation)
4. Review selected item(s) (Cart Page)
5. Proceed to checkout and review total (Checkout Page)
6. Choose a payment method — Cash, Touch 'n Go, or Card (Payment Method Selection)
7. Handle payment failure if it occurs (Payment Declined/Failed)
8. Receive confirmation of successful purchase (Order Confirmed)

This task breakdown corresponds directly to the proposed eight UI screens so that each design screen will support a step in the walk of the user to accomplish the task and not be some random addition to a UI.

5.0 Evaluation

The results of the survey are good quantitative evidence for the proposed redesign.

Satisfaction: 34.9% did rate 4-5 (easy), with most of them rating in the middle of 3 (**53.5%**) – the current interface is just "tolerable" not easy to use (**Figure 2**).

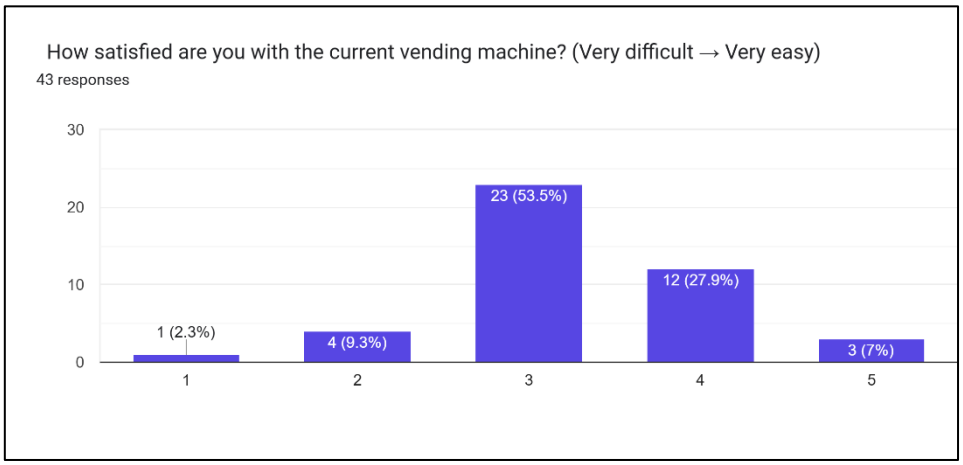


Figure 2: Student satisfaction ratings with the current vending machine (n = 43).

Payment confusion: 67.4% of students said they were at least once confused in the payment processes and features, which is why there is a need for steps to be given in the payment processes.

Demand for digital UI: 65.1% thoroughly agreed and **20.9%** agreed having a touchscreen interface to make it easier to use, meaning that **86%** said they would like it (**Figure 3**).

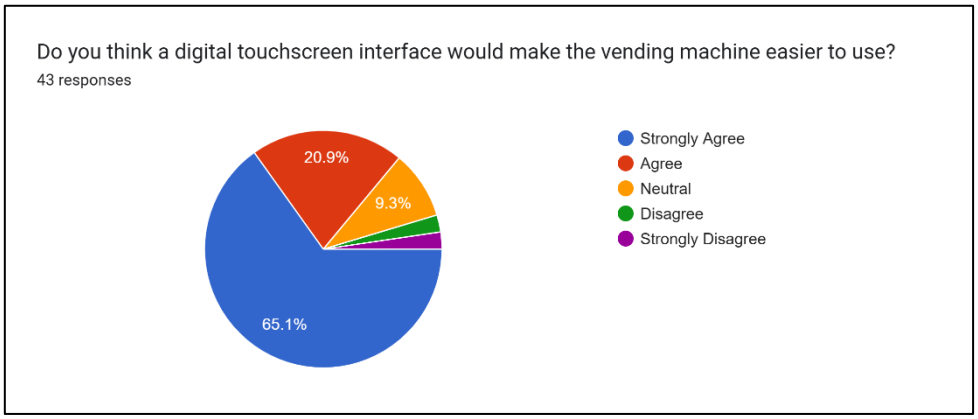


Figure 3: Student agreement that a digital touchscreen interface would improve usability.

Feature priority: Step-by-step instructions (**41.9%**) and digital payment information (**30.2%**) were the top-rated online functionality that would be most helpful (shown as a percentage; first two options) when asked which digital feature would help most (**Figure 4**). This directly informed the sequential, guided flow of the **8 proposed screens**.

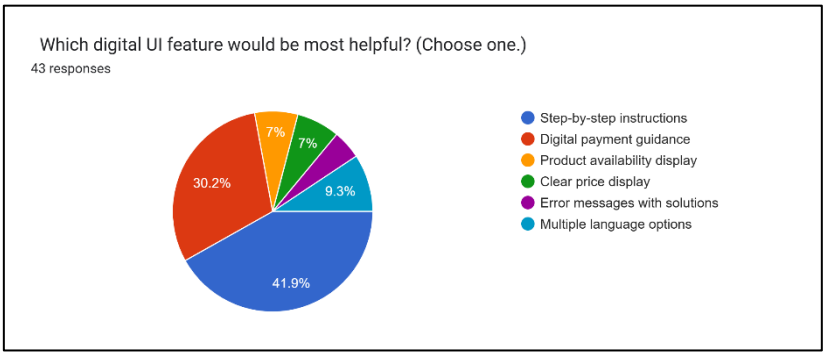


Figure 4: Most helpful digital UI feature as selected by students.

Overall impact: 83.7% agreed that having a digital UI would enhance the overall experience and **69.8%** said they would use it more frequently if it were digital (Figure 5).

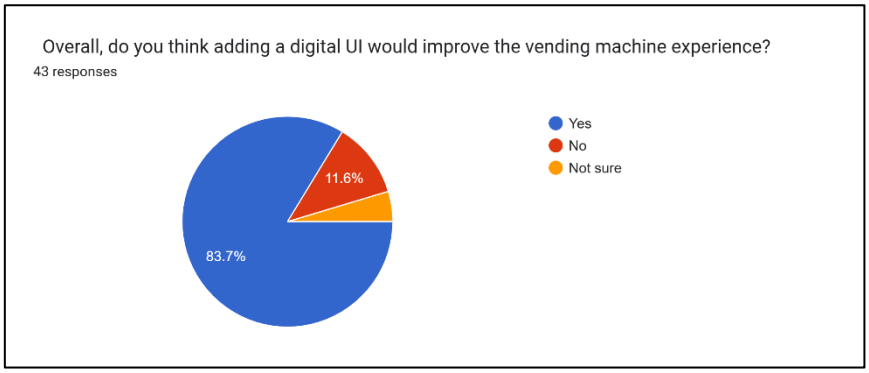


Figure 5: Overall student opinion on whether a digital UI would improve the experience.

6.0 Interaction Design

6.1 Prototypes

Eight screens were created in Figma that symbolized the whole purchase process, ranging from browsing to order confirmation.

Figma Vending Machine Project URL:
https://www.figma.com/design/sF264iTYbzw2D4oZBFAdCc/HCI-Vending-Machine-UI?node-id=0-1&p=f&t=3bXo7ffHJfffthdV-0

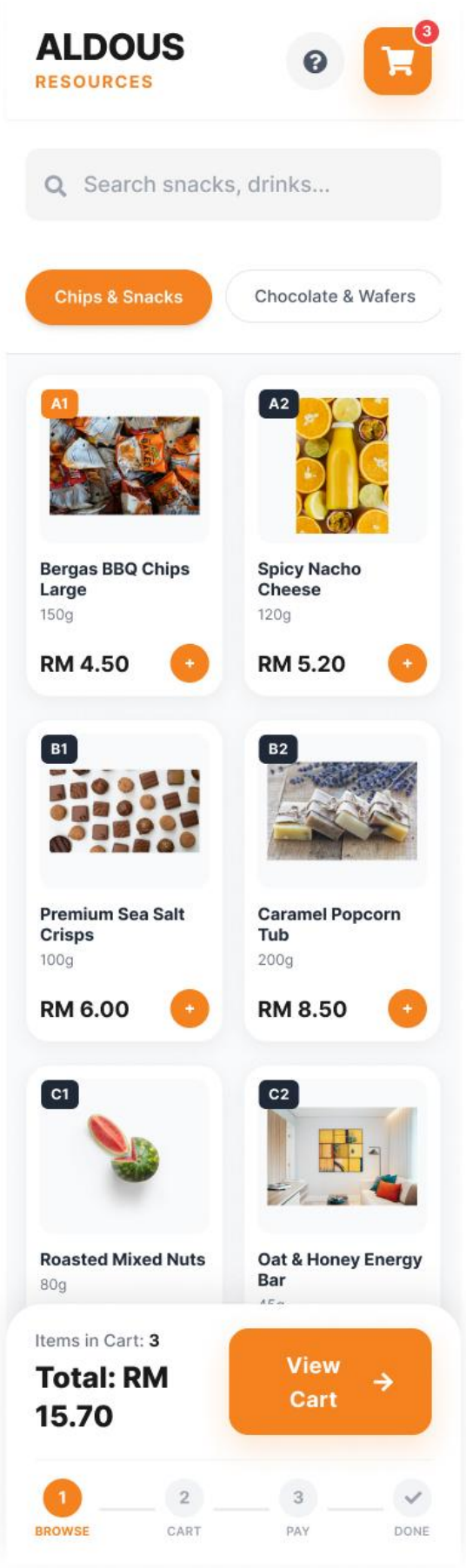


Figure 6: Main Dashboard — product browsing screen.

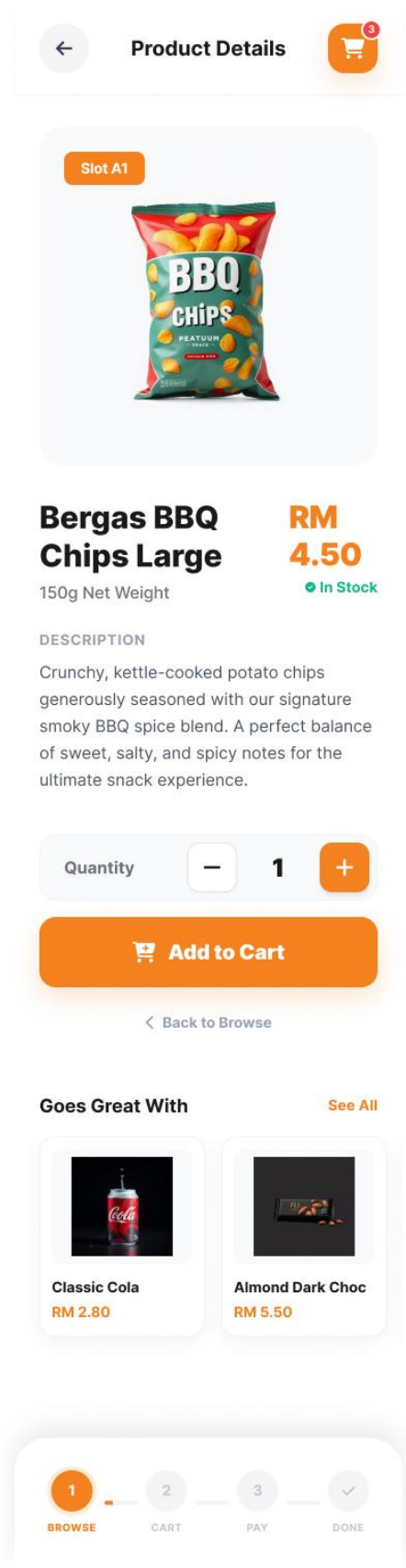


Figure 7: Product Select/Detail screen showing price and description.

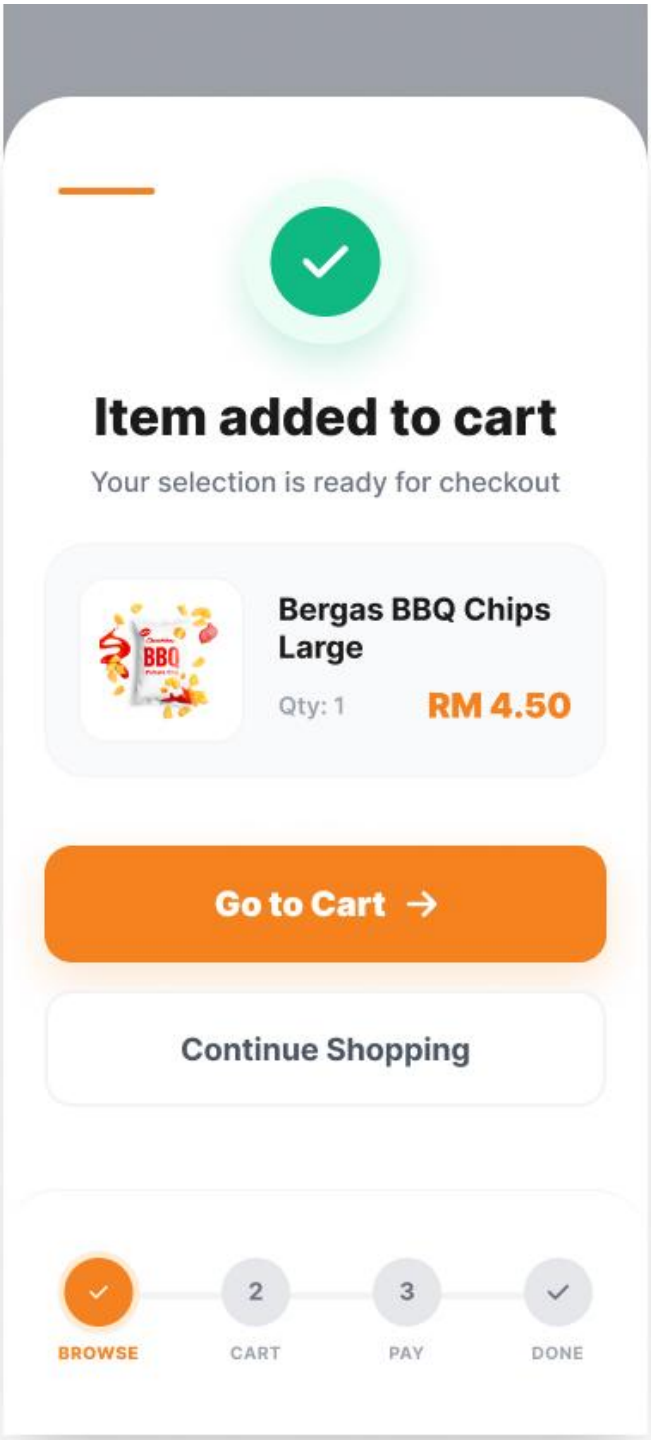


Figure 8: Add to Cart Confirmation screen.

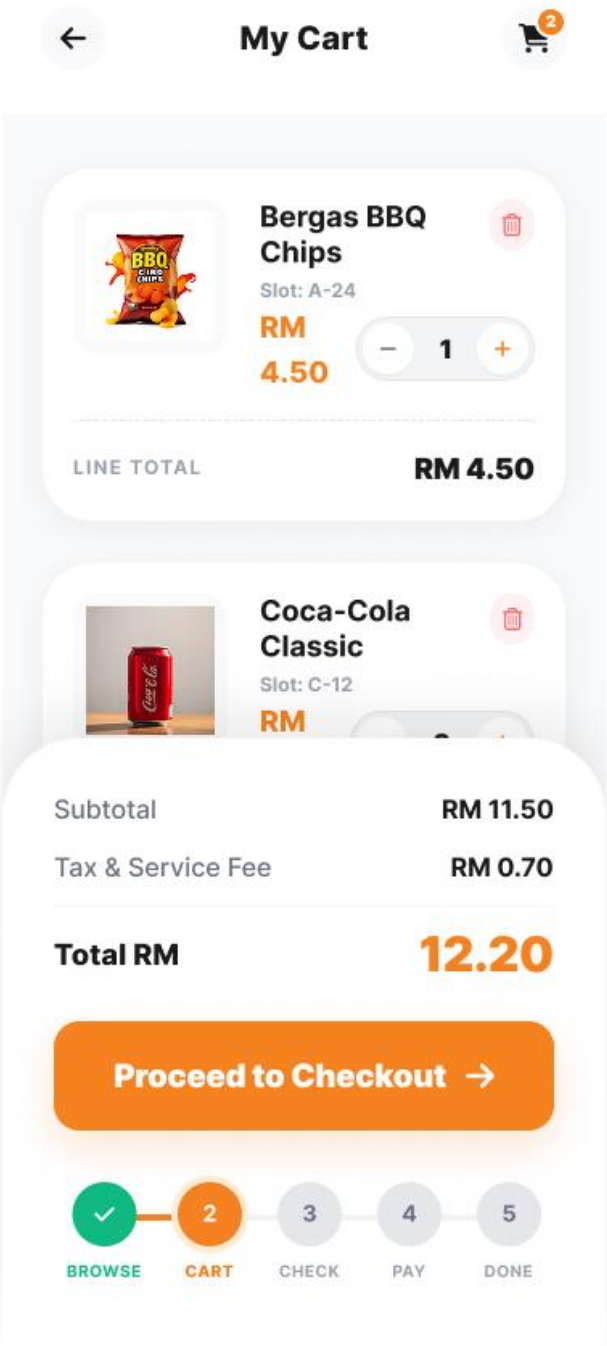


Figure 9: Cart Page showing selected items and total.

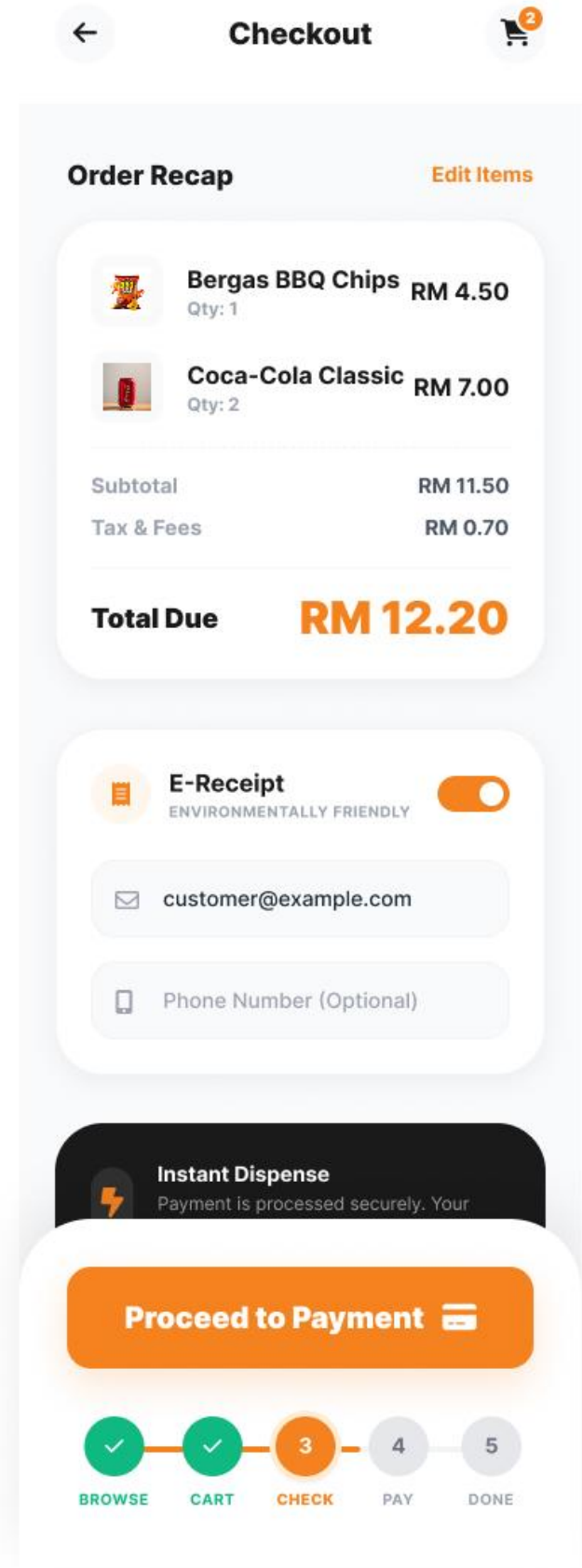


Figure 10: Checkout Page summarizing order before payment.

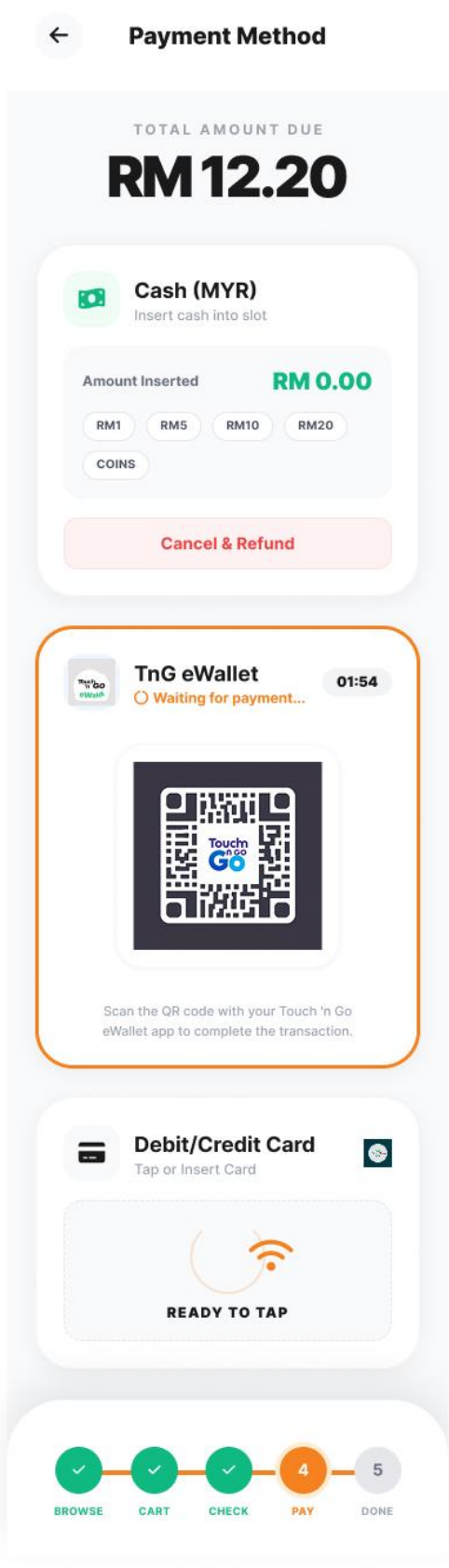


Figure 11: Payment Method Selection screen (Cash, Touch 'n Go, Card).

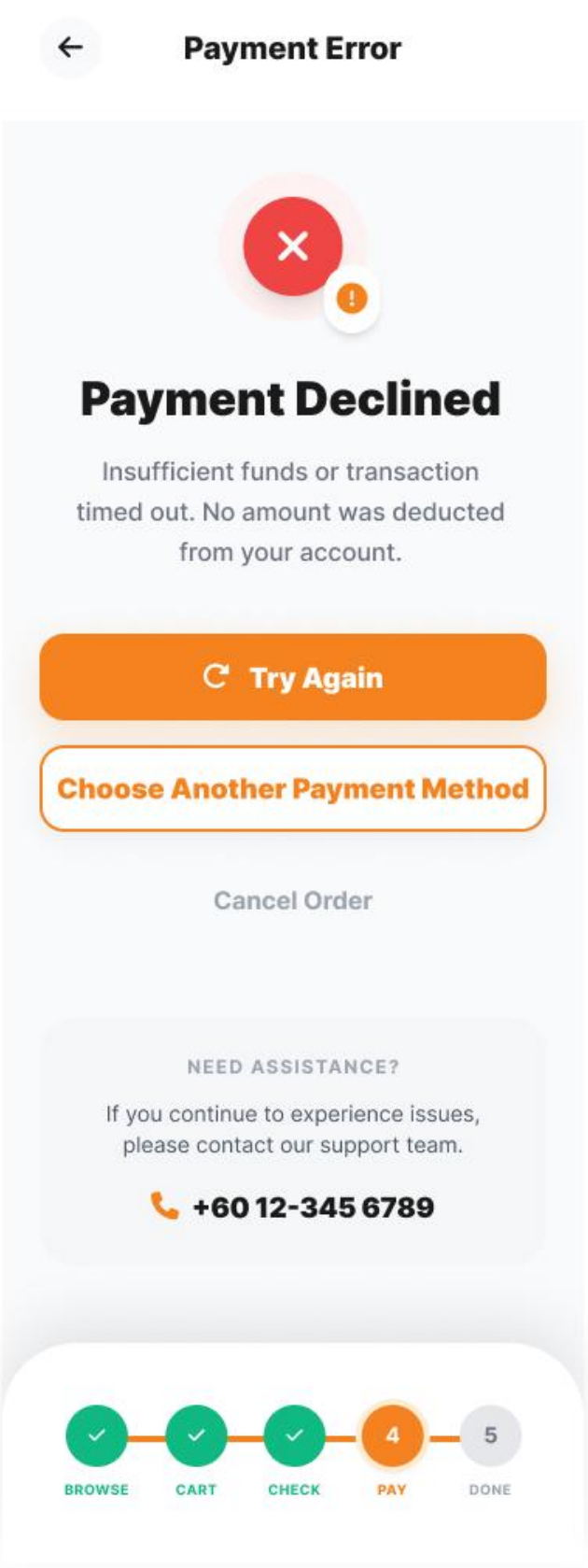


Figure 12: Payment Declined/Failed screen with retry option.

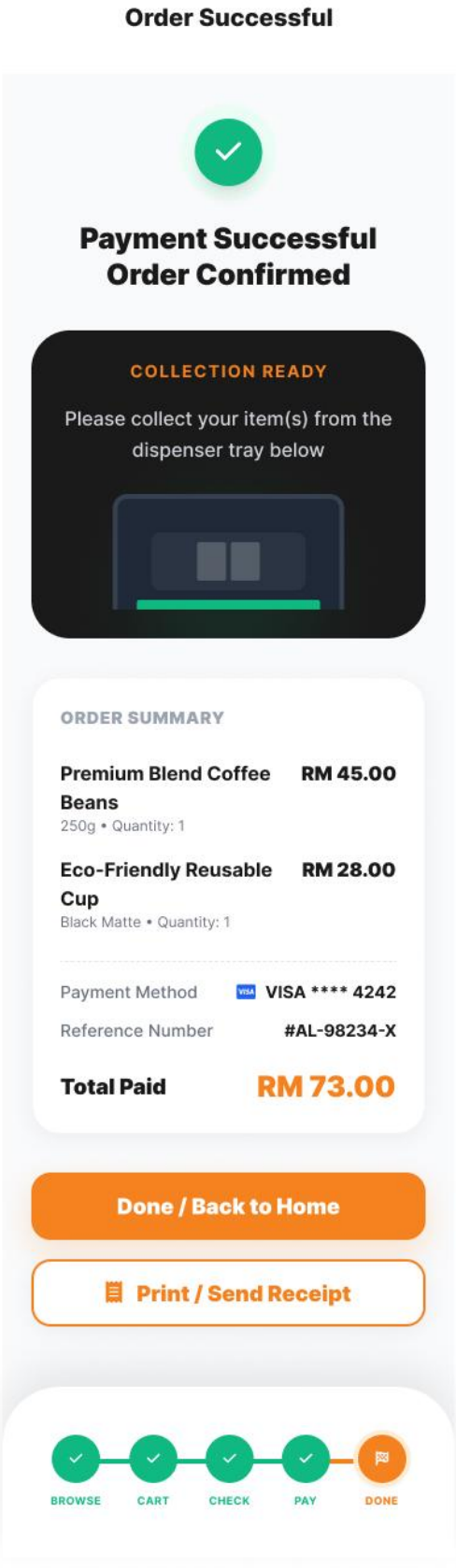


Figure 13: Order Confirmed screen with pickup instructions.

6.2 Justification on Prototype Design

All the screens discuss a specific issue from the survey. To overcome the **14%** segment who faced difficulty in checking product availability, and the price confusion, visual product and visual price clarity was provided via the Main Dashboard and Product Select screens. The Add to Cart Confirmation and Cart Page ensures that the user has control and prevention of any errors before accepting any purchase in accordance with Nielsen's (2020) Heuristic (User control and freedom).

67.4% who experienced payment confusion were directly met by the Payment Method Selection screen which provides clear and guided options such as cash (pay by Touch 'n Go) and card. The Payment Declined/Failed screen will provide visibility on the status of the system and recommended actions for recovery – which is not done in the current manual machine (Nielsen, 2020; Kang et al., 2023). Lastly, the Order Confirmed screen completes the interaction loop and relays to the person they have been promised that their transactions were successful, answering the complaint directly (**14%**) "product not dispensed".

The eight screens reflect a linear, low-cognitive-load progression that corresponds with the step-by-step guidance, which students identified as the most important (**41.9%, Section 5.0**).

7.0 Conclusion and References

Conclusion

The survey results indicated that made **60.5%** of students encountered problems using the existing manual vending machine at INTI International University, with the majority of them reporting payment confusions (**53.5%**) and unclear instructions (**14%**). In turn, an **8-screen digital touchscreen UI** was designed and validated through survey results and the well-known principles of **HCI**. **83.7%** of the students think that the use of a digital UI would make the vending more usable, more accessible and more efficient, this proposed redesign would directly tackle real needs of the users with a guided navigation through the user interface, clear availability of payment methods and clear management of errors.

References

1. Aljohi, A.S., Alzaabi, S.S., Almahri, R.S., Tsaramirsis, G. and Hamid, O.H. (2023) 'Navigating the Divide: Digital Kiosks and Mobile Apps as Complementary Human-Centered Self-Service Technologies', Engineering Proceedings, 59(1), p. 162. Available at: https://doi.org/10.3390/engproc2023059162 (Accessed: 17 July 2026).
2. Ayob, M.A. and Baba, N. (2025) 'Self-ordering kiosk usage and purchase behavior in quick service restaurants: A case in Alor Setar's quick service restaurants', International Journal of Research and Innovation in Social Science, 9(15), pp. 779–789. Available at: https://doi.org/10.47772/IJRISS.2025.915EC0056 (Accessed: 17 July 2026).
3. Kang, S., Kim, J. and Lee, S. (2023) 'Comparative Analysis of Usability and Accessibility of Kiosks for People with Disabilities', Applied Sciences, 13(5), p. 3058. Available at: https://doi.org/10.3390/app13053058 (Accessed: 17 July 2026).
4. Nielsen, J. (2020) 10 Usability Heuristics for User Interface Design. Nielsen Norman Group. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/ (Accessed: 17 July 2026).

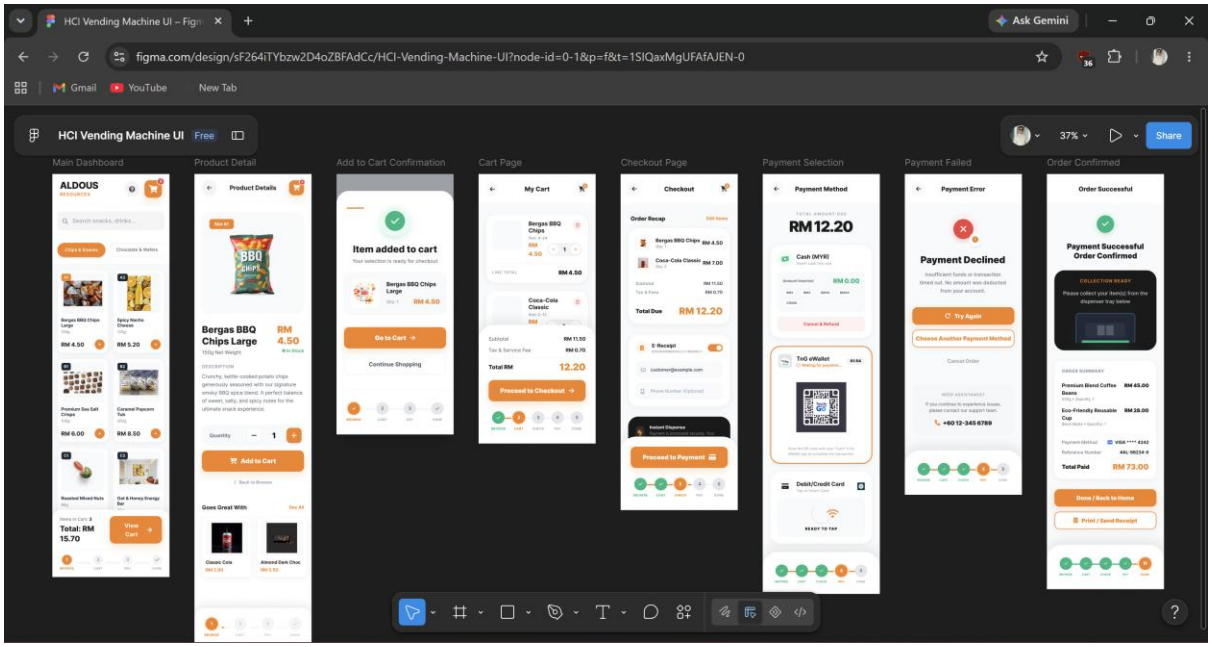
Appendices:

Group Contribution:

Type of Work	Shahadat Hossain I26041279	Rowan Cesar Macuacua I25037217	Raduan Mia I25035605	Morris Kinyua I24028729	Total
1. Report Writing	100%	0%	0%	0%	= 100%
2. Google Form Survey & Data Collection	0%	33.3%	33.3%	33.3%	= 100%
3. Data Analysis (Charts & Findings)	100%	0%	0%	0%	= 100%
4. Figma UI Design (8 Screens)	100%	0%	0%	0%	= 100%
5. Research & References	100%	0%	0%	0%	= 100%
6. Formatting, Editing & Compilation	100%	0%	0%	0%	= 100%

Figma Vending Machine Project URL:

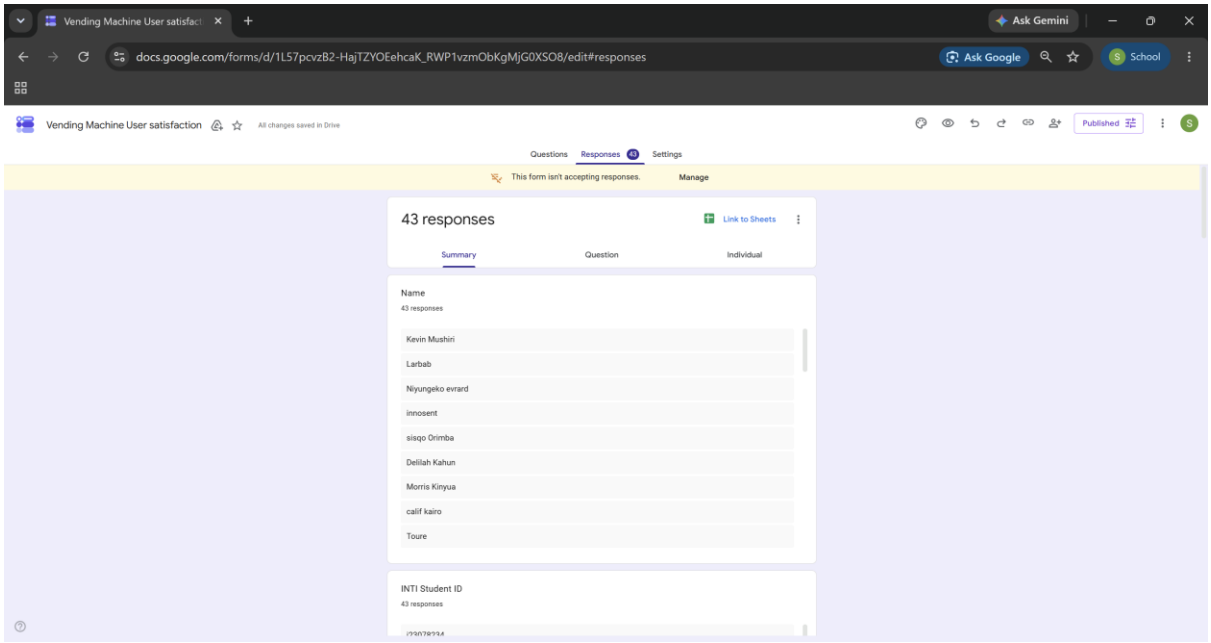
https://www.figma.com/design/sF264iTYbzw2D4oZBFAdCc/HCI-Vending-Machine-UI?node-id=0-1&p=f&t=3bXo7ffHJfffthdV-0

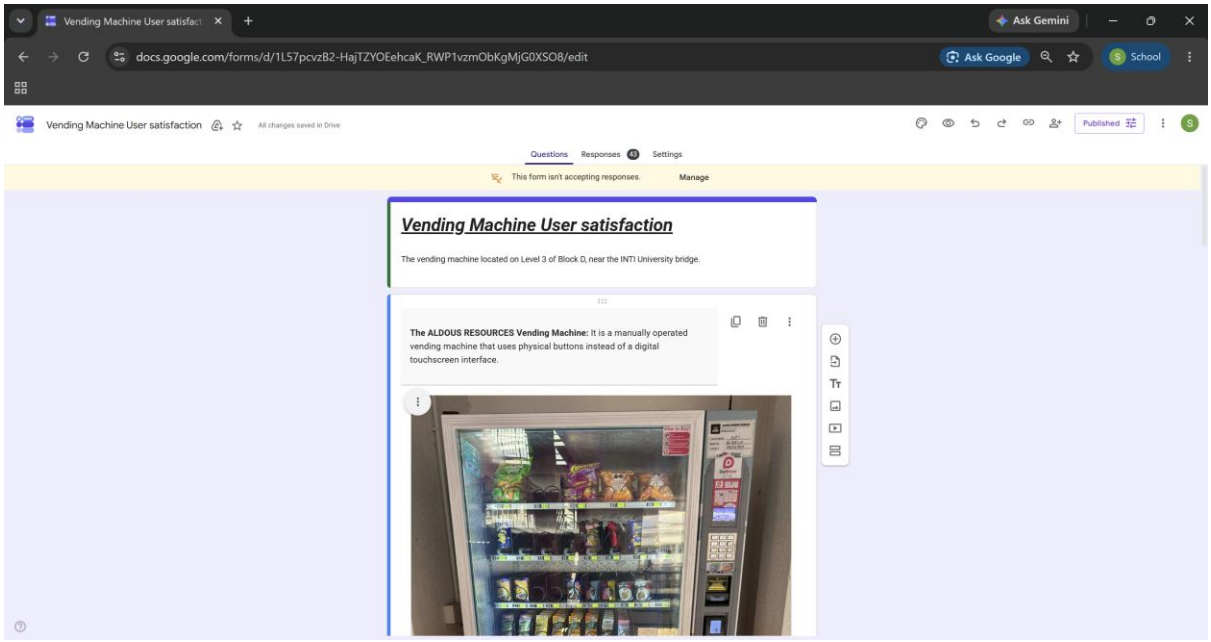


The **ALDOUS RESOURCES** Vending Machine Real Image:



Google Form Survey Screenshots:

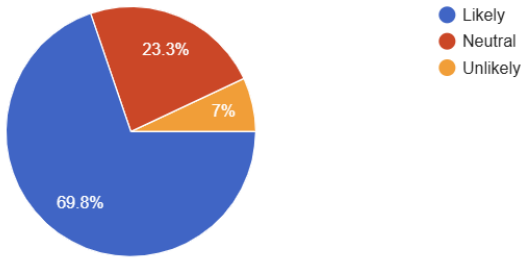




If this vending machine had a digital UI, how likely would you be to use it more often?

Copy chart

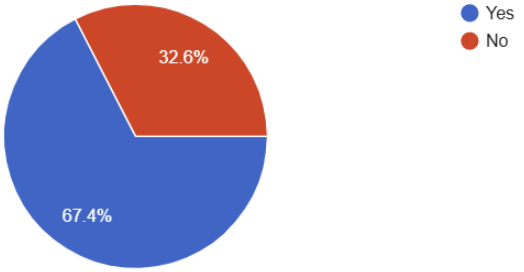
43 responses

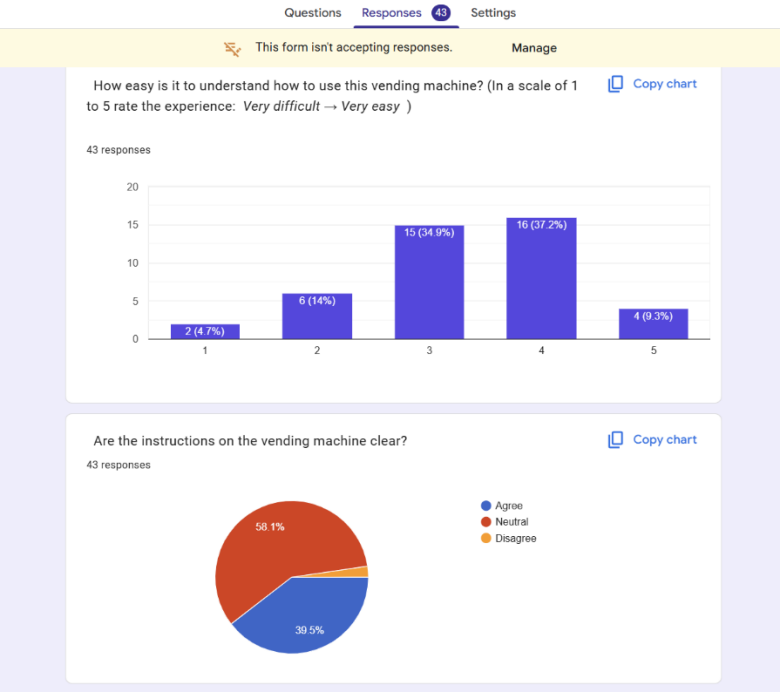


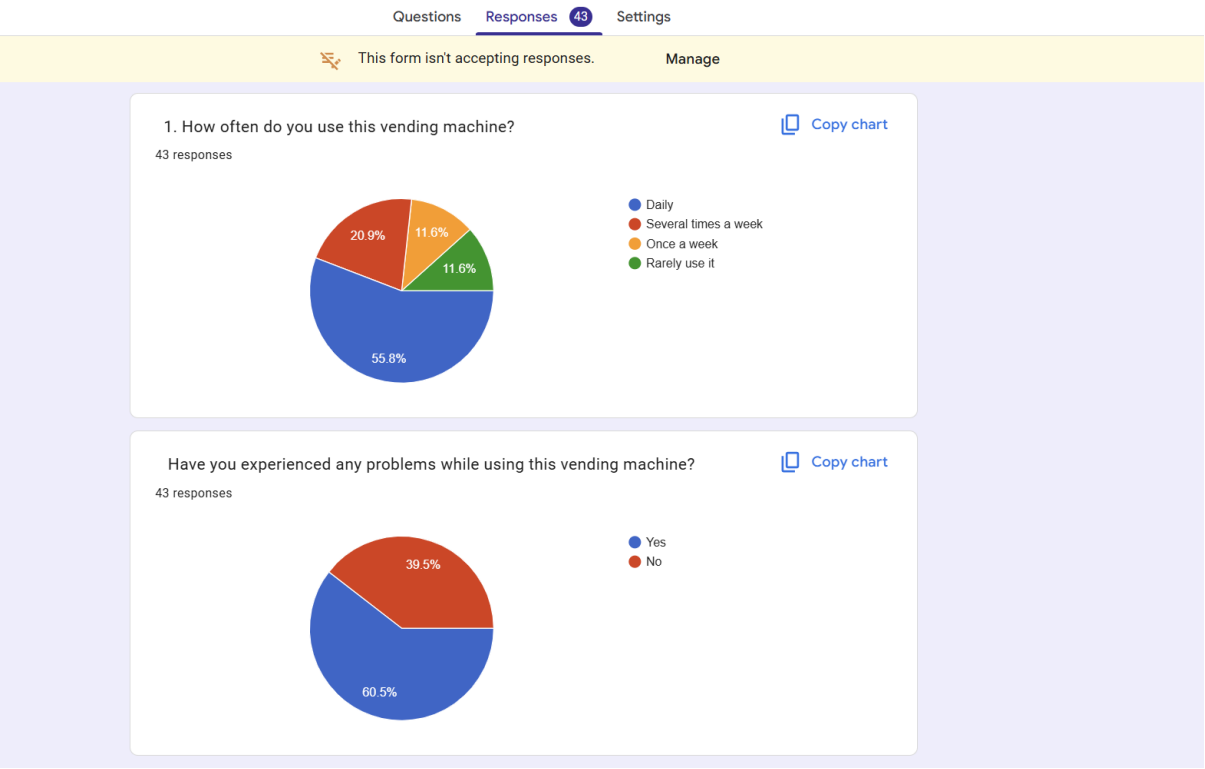
Copy chart

Have you ever been confused about the payment process?

43 responses




Questions
Responses 43
Settings
This form isn't accepting responses.
Manage
How easy is it to understand how to use this vending machine? (In a scale of 1 to 5 rate the experience: Very difficult → Very easy)
Copy chart
43 responses
20
15
10
5
0
2 (4.7%)
6 (14%)
15 (34.9%)
16 (37.2%)
4 (9.3%)
1
2
3
4
5
Are the instructions on the vending machine clear?
Copy chart
43 responses
58.1%
39.5%
Agree
Neutral
Disagree


Questions
Responses 43
Settings
This form isn't accepting responses.
Manage
1. How often do you use this vending machine?
Copy chart
43 responses
20.9%
11.6%
11.6%
55.8%
Daily
Several times a week
Once a week
Rarely use it
Have you experienced any problems while using this vending machine?
Copy chart
43 responses
39.5%
60.5%
Yes
No